

Summary On A Page

All developed models outperformed the benchmark in predicting loan repayment failure based on the Gini score (a measure of model accuracy). The selected model was chosen to balance accuracy and simplicity. It has an accuracy measure of 37% compared to the benchmark's 11% on unseen data, and it shares annual income and employment length with significant variables from the literature.

Statically significant Variables:

Interest Rate: Every 1% increase in the interest rate increases the odds of loan repayment failure by **17%** (with a 95% uncertainty ranging from **14% to 20% increase**).

Inquiries in the Last 6 Months: Each additional inquiry raises the risk of failure by **16%** (uncertainty: **13% to 18% increase**).

Annual Income: a \$10,000 increase in income decreases failure by **10%** (**7% to 13%**).

Loan Term: Longer loans (60 months) reduce failure risk by **97%** (**95% to 98%**), while shorter loans (36 months) reduce it by **98%** (**97% to 98%**).

Revolving Utilisation Rate: A 1% increase in utilisation increases failure risk by **1%** (**1% to 2%**).

Employment length of 10+ years: Loan applicants with employment length of 10+ years increase risk by **14%** (**3.8% to 24.6%**).

Loan Amount: A \$10,000 increase in the loan amount increases risk by **16%** (**0.04% to 34%**).

Statically significant Interactions Between Variables:

Interest Rate & Revolving Utilisation: This interaction slightly decreases the failure risk by **0.1%** (uncertainty: **0.04% to 0.12% decrease**).

Annual Income & Revolving Utilisation: A slight risk increase of **0.06%** (**0.02% to 0.1%**).

Accounting for Variation Between States and Issue Dates of Loans

Following on from the Initial Model, we assessed the effect of accounting for the variation between states where customers live and when they received their loans. The results from this analysis indicated that the mixed effects model did improve the performance of the original credit risk model.

The original model's Gini Score was 0.384 and 0.369 for the training and validation sets respectively, while the Gini Score was 0.422 for the mixed effects model including both zip code and issue dates as random effects.

While other metrics indicated that other models developed were potentially better at explaining the variation while being less complex, the improvements found in the metrics mentioned above indicate that the model accounting for the variation found between states and when the loans were issued will be able to better distinguish customers who can pay off their loans from those who cannot.

Loans of 2007 and 2008 had a negative effect on loan repayment.

The states of NV (Nevada), FL (Florida), CA (California) and GA (Georgia) had a negative effect on loan repayment, whereas CO (Colorado) and OH (Ohio) had a positive one.